

EE2027 Midterm Cheat Sheet

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Circuit Analysis Fundamentals

Basic Laws & Theorems

- **KCL (Node Analysis):** $\sum I_{in} = 0$.
- **KVL (Mesh Analysis):** $\sum V_{loop} = 0$.
- **Superposition:** Analyze one independent source at a time. Nullify others: Voltage source $\rightarrow$ Short; Current source $\rightarrow$ Open.
- **Thevenin Equivalent:** (V_{TH} in series with R_{TH})
 - **To find V_{TH} :** Remove load, find open-circuit voltage V_{oc} across terminals.
 - **To find R_{TH} :**
 - * Turn off independent sources (V-source $\rightarrow$ short, I-source $\rightarrow$ open).
 - * Find equivalent resistance R_{eq} from terminals.
 - * If dependent sources exist: Use a test source. $R_{TH} = V_{test}/I_{test}$.
- **Norton Equivalent:** (I_N in parallel with R_N)
 - **To find I_N :** Short the terminals, find short-circuit current I_{sc} .
 - **To find R_N :** Same as finding R_{TH} . So, $R_N = R_{TH}$.
- **Source Transformation:** $V_{TH} = I_N \times R_{TH}$.
- **Max Power Transfer:** Occurs when $R_L = R_{TH}$.
 $P_{max} = \frac{V_{TH}^2}{4R_{TH}}$.

AC Signals & Impedance

- **Phasor:** $v(t) = V_p \cos(\omega t + \theta) \Leftrightarrow \mathbf{V} = V_{rms} \angle \theta$, where $V_{rms} = V_p / \sqrt{2}$ for sinusoids.
- **Impedance (Z):** $Z_R = R$, $Z_C = \frac{1}{j\omega C}$, $Z_L = j\omega L$.
- **Passive RC Filters (1st Order):**
 - LPF: $H(j\omega) = \frac{1}{1+j\omega RC}$
 - HPF: $H(j\omega) = \frac{j\omega RC}{1+j\omega RC}$
 - 3dB frequency for both: $f_{3dB} = \frac{1}{2\pi RC}$.

PN Junction Diode

IV Characteristic & Models

Diode Equation: $I_D = I_S (e^{\frac{V_D}{nV_T}} - 1)$, V_T is 0.025

$$r_d = \frac{nV_T}{I_D}, R = \frac{V_{DD} - V_D}{I_D}$$

$$\text{RMS} = i_{d,rms} = \frac{v_{dd,rms}}{R + r_d}$$

- **Forward Bias ($V_D > 0$):** Current flows easily.
- **Reverse Bias ($V_D < 0$):** Acts as open circuit, $I_D \approx -I_S \approx 0$.

- **Breakdown ($V_D < -V_Z$):** Voltage clamps at V_Z .

DC (Large-Signal) Models:

- **CVD:** Replace with ideal diode + 0.7V source. OFF if voltage is $< 0.7V$, ON (acts as 0.7V source) if current tries to flow.
- **Piecewise-Linear:** Ideal diode + V_{DO} source + series resistance r_D .

AC (Small-Signal) Model:

- Find DC current I_D , then replace diode with resistance r_d .
- $r_d = \frac{nV_T}{I_D}$ where $V_T \approx 25.9mV$ at 300K.
- $\mathbf{v_o} = \frac{r_d}{r_d + R_s} = \frac{nV_T}{nV_T + IR_s} \mathbf{v_s}$ voltage output from the input

Diode Applications

- **Rectifiers:** Half-wave and full-wave (bridge) circuits to convert AC to DC.
- **Zener Regulator:** Used in reverse breakdown to provide a stable voltage V_Z .

Bipolar Junction Transistor (BJT)

Modes of Operation (NPN)

- **Forward Active:** BE Forward, BC Reverse (Amplifier).
- **Saturation:** BE Forward, BC Forward (Closed Switch).
- **Cut-off:** BE Reverse, BC Reverse (Open Switch).

DC Analysis (Forward Active)

Replace BJT with its DC model to find Q-point (I_C, V_{CE}).

- **BE Junction:** Diode model, assume $V_{BE} \approx 0.7V$.
- **Current Relations:** $I_C = \beta I_B$, $I_E = (\beta + 1)I_B$.

AC Analysis (Small-Signal)

1. Find DC Q-point I_C . 2. Calculate parameters:

- **Transconductance:** $g_m = \frac{I_C}{V_T}$
- **Input Resistance:** $r_\pi = \frac{\beta}{g_m}$
- **Output Resistance:** $r_o = \frac{V_A}{I_C}$ (if Early Voltage V_A given)

3. Replace BJT with hybrid- π model and solve AC circuit.

Operational Amplifiers (Op-Amps)

Gain = 100, also mean $20 \times \log_{10} 100 = 40dB$

Ideal Op-Amp Rules (with Neg. Feedback)

1. **Virtual Short:** $v_- = v_+$
2. **Zero Input Current:** $i_- = i_+ = 0$

Key Configurations

- **Inverting Amp:** $V_{out} = -\frac{R_f}{R_1} V_{in}$
- **Non-inverting Amp:** $V_{out} = (1 + \frac{R_f}{R_1}) V_{in}$
- **Voltage Follower (Buffer):** $V_{out} = V_{in}$
- **Summing Amp:** $V_{out} = -R_f (\frac{V_1}{R_1} + \frac{V_2}{R_2} + \dots)$
- **Difference/Instrumentation Amp:** $V_{out} = \frac{R_2}{R_1} (V_2 - V_1)$. Rejects common-mode signals.
- **Integrator:** $V_{out}(t) = -\frac{1}{RC} \int V_{in}(t) dt$
- **Differentiator:** $V_{out}(t) = -RC \frac{dV_{in}(t)}{dt}$
- **Superdiode:** Overcomes V_D drop. For $V_{in} > 0$, $V_{out} \approx V_{in}$.
- **Comparator (No Feedback):** V_{out} swings to $+V_{sat}$ or $-V_{sat}$.

Active Filters

- **1st Order LPF:** $H(j\omega) = -\frac{R_f}{R_1} \frac{1}{1+j\omega R_f C}$
- **1st Order HPF:** $H(j\omega) = -\frac{R_f}{R_1} \frac{j\omega R_1 C}{1+j\omega R_1 C}$
- **2nd Order Sallen-Key LPF:** $H(s) = \frac{K\omega_o^2}{s^2 + s(\frac{\omega_o}{Q}) + \omega_o^2}$ where $\omega_o = \frac{1}{RC}$ and $\frac{1}{Q} = 3 - K$.
- **2nd Order Sallen-Key HPF:** $H(s) = \frac{Ks^2}{s^2 + s(\frac{\omega_o}{Q}) + \omega_o^2}$ with same ω_o, Q definitions.

Non-Ideal Op-Amp Effects

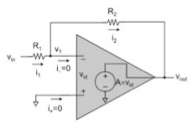
- **GBW:** Bandwidth is where loop gain $|\beta A(j\omega)| = 1$.

- Non-inv: $f_{3dB,CL} = \frac{GBW}{1+R_f/R_1}$
- Inv: $f_{3dB,CL} = GBW \times \frac{R_1}{R_1+R_f}$

- **Slew Rate (SR):** Max output slope. For a sine wave $V_p \sin(2\pi ft)$, need $SR \geq 2\pi f V_p$.
- **Input Offset (V_{OS}):** A DC error voltage. Output DC Error: $V_{out,OS} = V_{OS} \times (1 + \frac{R_f}{R_1})$. Reduces available output swing.
- **Input Noise (v_n):** Output RMS noise $v_{out,n} = (\text{Noise Gain}) \times v_n \times \sqrt{ENBW}$. For 1st-order filter, $ENBW = f_{3dB,CL} \times \frac{\pi}{2}$.
- **CMRR:** Rejection of common-mode signals. Ideal is ∞ .
- **PSRR:** Rejection of power supply noise. Ideal is ∞ .

Analysis Note: For noise (v_n) and offset (V_{OS}), model as a voltage source on the non-inverting (+) input and amplify by the non-inverting gain ("noise gain," $1 + R_f/R_1$). Equivalent Noise Bandwidth (ENBW), $ENBW = f_{3dB} \times \frac{2}{\pi}$ for a first-order filter.

OP-AMPS



In circuit Analysis
 $V_- = 0$
 $V_+ = 0$
 $i_- = i_+$
 $V_- = V_+$ Virtual short
 $R_{in} = \infty$
 $R_{out} = 0$
 $A = \infty$

Open Loop (SSA)
 - Difficult to control gain
 - Simple
 - Low Power consumption
 - Poor Linearity / High Distortion

Feedback (Op-Amp) Amplifiers.
 - Accurate gain from Resistor
 - Complicated
 - High Power consumption
 - High Linearity / Low Distortion

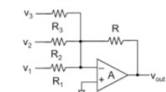
VS

1. Inverting



$$V_{out} = -\frac{R_2}{R_1} V_{in}$$

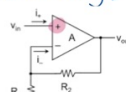
2. Summing



Superposition

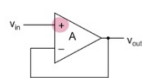
$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

3. Non-Inverting



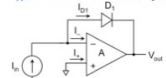
$$V_{out} = \left(1 + \frac{R_2}{R_1} \right) V_{in}$$

4. Source Follower



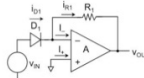
$$V_{out} = V_{in}$$

5. Logarithmic



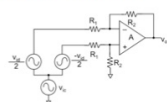
$$V_{out} = -R_f \ln \left(\frac{I_{in}}{I_s} \right)$$

6. Exponential



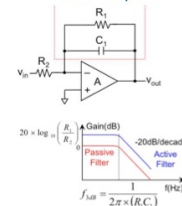
$$V_{out} = -R_f I_s e^{V_{in}/V_T}$$

7. Instrumentation



$V_{out} = \frac{R_2}{R_1} (V_{id})$
 Rejects Common mode.
 Amplifies differential
 Good for blocking noise
 Used in measuring instrument

10. 1st Order Lowpass



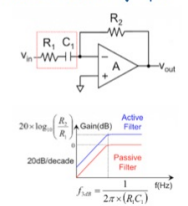
$$f_{LP} = \frac{1}{2\pi \times (R_2 C_1)}$$

8. Integrator



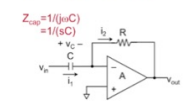
$V_{out}(t) = -\frac{1}{RC} \int_0^t V_{in}(t) dt$
 $H(j\omega) = -\frac{Z_{cap}}{R}$
 Gain ↓ as Freq ↑

11. 1st Order High pass



$$f_{LP} = \frac{1}{2\pi \times (R_1 C_1)}$$

9. Differentiator



$$V_{out}(t) = -RC \frac{d}{dt} V_{in}(t)$$

Op Amp Parameters

Open Loop Voltage Gain (A_{OL})

How well the desired signal is amplified

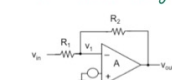
Common Mode Voltage Gain (A_{CM})

How much noise is amplified Ideal Value is 0

Common Mode Rejection Ratio (CMRR)

$CMRR [dB] = 20 \log \left(\frac{A_{OL}}{A_{CM}} \right)$ How well noise is rejected Ideal is ∞

Input Offset Voltage (V_{OS})



DC output shift
 Use superposition to find a new V_{out}

Power Supply Rejection Ratio (PSRR)



How well supply line ripples are rejected
 Ideal is ∞

$$A_{sup} = \frac{V_{O, sup}}{V_{S, sup}}$$

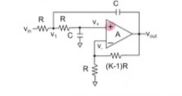
$$PSRR [dB] = 20 \log \left(\frac{A_{OL}}{A_{sup}} \right)$$

Slew Rate (SR)

$$SR = 2\pi f \cdot V_{out, pk} \cdot \sin(2\pi f t)$$

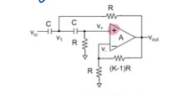
Maximum slope that can occur at output

13. 2nd Order Sallen-Key Lowpass Filter



$$f_{LP} = \frac{1}{2\pi \times (R_1 R_2 C_1 C_2)}$$

14. 2nd Order Sallen-Key High Pass Filter



$$f_{LP} = \frac{1}{2\pi \times (R_1 R_2 C_1 C_2)}$$

$\omega_0 = \frac{1}{RC} \left| \frac{1}{Q} = 3 - K \right| H_0 = K$
 at Low Freq $\rightarrow H(j\omega) \approx H_0$
 at High Freq $\rightarrow H(j\omega) \approx \frac{H_0 \omega_0^2}{(\omega)^2}$

at Low Freq $\rightarrow H(j\omega) \approx \frac{H_0 \omega_0^2}{\omega^2}$
 at High Freq $\rightarrow H(j\omega) \approx H_0$

15. Superdiode Rectifier

